



MAT110

Differential Calculus and Coordinate Geometry

Assignment 03

Spring 24

Submission Date: 17th April, 2024 (Wednesday)

Instructions:

Assignment should be handwritten. Please write your Name and ID on the cover page of your assignment answer script. Make PDF of your assignment and submit it to the provided google form.

Solve all problems. Answer the questions by yourself. Plagiarism will lead to an F grade in the course. Each question carries 10 marks. Total marks is 60 and it will be converted to 15.

1. Locate all relative maxima, relative minima and saddle points, if any

$$f(x, y) = xy + \frac{2}{x} + \frac{4}{y}$$

2. Use Lagrange multipliers to find the maximum and minimum values of $f(x, y, z) = 2x + y - 2z$, subject to the given constraint $x^2 + y^2 + z^2 = 4$. Also, find the points at which these extreme values occur.

3. Use appropriate forms of the chain rule to find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$

$$z = e^{x^2y}; \quad x = \sqrt{uv}, \quad y = \frac{1}{u}$$

4. Find the directional derivative of $f(x, y, z)$ at P in the direction of $\mathbf{a}$

$$f(x, y, z) = y - \sqrt{x^2 + z^2}; \quad P(-3, 1, 4); \quad \mathbf{a} = 2\mathbf{i} - 2\mathbf{j} - \mathbf{k}$$

5. Let, $f(x, y) = x^3 \ln(yx^2) + x^4y - e^{3x}x^2$. Find $f_{xy}, f_{xx}, f_{yy}, f_{yxx}$.

6. Suppose $\mathbf{A} = 2yz\mathbf{i} - x^2y\mathbf{j} + xz^2\mathbf{k}$ and $\phi = 2x^2yz^3$. Find $(\mathbf{A} \times \nabla)\phi$.